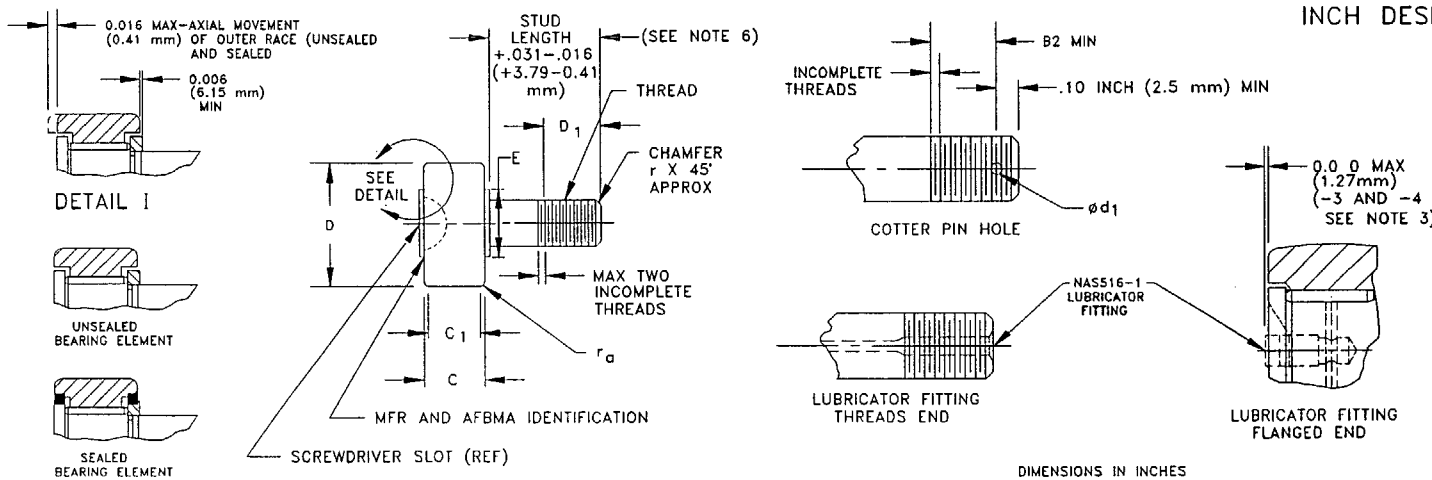
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INCH DESIGN



(B) DENOTES CHANGES

DIMENSIONS IN INCHES

FIRST DASH NO.	ød STUD	øD OUTER RING OUTSIDE DIA	C OUTER RING WIDTH	C_1 TRACK CONTACT WIDTH MIN	THREAD SIZE UNJF-3A	B_1 THREAD LENGTH REF.	$ød_a$ WASHER BACK-UP MIN.	B_2 MIN.	$ød_1$ COTTER PIN MALE	r_a 1/ FILLET MAX	r	TOTAL RADIAL PLAY MAX	LIMIT LOAD RATING lbf	2/ lb-in MAX	TRACK CAPACITY 40 C lbf	LOAD RATING A a TRACK ROLLER lbf	MASS (APPROX) lb
-3	.1900	.5000	.281	.230	.1900-32	.344	.297	.211	.070	.010	.031	.0017	790	8	385	395	.014 +(GRIPLENGTH NO. X .0005)
-4	.2500	.6875	.281	.230	.2500-28	.344	.359	.224	.076	.010	.031	.0017	940	20	525	470	.031 +(GRIPLENGTH NO. X .0009)
-5	.3125	.7500	.344	.290	.3125-24	.359	.422	.234	.076	.010	.047	.0017	1660	30	725	830	.043 +(GRIPLENGTH NO. X .0014)
-6	.3750	.8750	.469	.380	.3750-24	.359	.500	.265	.106	.025	.047	.0017	2720	55	1100	1360	.0 1 +(GRIPLENGTH NO. X .0020)
-7	.4375	1.0000	.531	.430	.4375-20	.422	.562	.283	.106	.025	.047	.0017	3860	150	1425	1930	.125 +(GRIPLENGTH NO. X .0026)
-8	.5000	1.1250	.656	.530	.5000-20	.422	.625	.314	.106	.040	.047	.0017	6040	205	1925	3040	.190 +(GRIPLENGTH NO. X .0035)

DIMENSIONS IN MILLIMETERS

FIRST DASH NO.	ød STUD	øD OUTER RING OUTSIDE DIA	C OUTER RING WIDTH	C_1 TRACK CONTACT WIDTH MIN	THREAD SIZE UNJF-3A	B_1 THREAD LENGTH REF.	$ød_a$ WASHER BACK-UP MIN.	B_2 MIN.	$ød_1$ COTTER PIN MALE	r_a 1/ FILLET MAX	r	TOTAL RADIAL PLAY MAX	LIMIT LOAD RATING	2/ MAX	TRACK CAPACITY 40 C N	LOAD RATING A a TRACK ROLLER	MASS (APPROX) kg
-3	4.826	12.700	7.14	5. 4	.1900-32	8.7	7.54	5.4	1.8	.2	.8	.043	3510	1.1	1710	1760	.006 +(GRIPLENGTH NO. X .0002)
-4	6.350	17.462	7.14	5. 4	.2500-28	8.7	9.12	5.7	1.9	.25	.8	.043	4100	2.3	2340	2090	.014 +(GRIPLENGTH NO. X .0004)
-5	7.925	19.050	8.74	7.37	.3125-24	9.1	10.72	5.9	1.9	.25	1.2	.043	7300	3.4	3220	3690	.020 +(GRIPLENGTH NO. X .0006)
-6	9.525	22.225	11.91	9.65	.3750-24	9.1	12.70	6.7	2.7	.64	1.2	.043	12100	6.0	4890	6050	.037 +(GRIPLENGTH NO. X .0009)
-7	11.100	25.400	13.49	10.92	.4375-20	10.7	14.27	7.2	2.7	.64	1.2	.043	17200	7.0	6340	8500	.057 +(GRIPLENGTH NO. X .0012)
-8	12.700	2 .575	16.66	13.46	.5000-20	10.7	15.	8.0	2.7	.02	1.2	.043	27000	23.0	87 0	13500	.0 6 +(GRIPLENGTH NO. X .0016)

1/ THE CHAMFER ON BEARINGS MUST CLEAR THE MAXIMUM FILLET RADIUS GIVEN IN THE TABLE. THIS SPECIFICATION DOES NOT CONTROL BEARING CHAMFER CONTOURE.

2/ INSTALLATION TORQUE LUBRICATED THREADS.

Form Approved OMB No. 0704-0188

PREPARING ACTIVITY: NAVY-AS

CUSTODIANS: ARMY- AT NAVY- AS
AIR FORCE- 99 DLA-

REVIEW: AF-84 DLA-IS

USER:

PROJECT NUMBER: 3110-0909

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MILITARY SPECIFICATION SHEET

TITLE

BEARING, ROLLER, NEEDLE, TRACK ROLLER, INTEGRAL STUD, TYPE VII, ANTIFRICTION, INCH

SPECIFICATION SHEET NUMBER

MS21432 05 DEC 94 REV B

SUPERSEDING MS21432A 21 MARCH 1977

AMSC- N/A FSC 3110

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TABLE I. TOLERANCE VALUE

DIMENSIONS IN INCHES

Ød BASIC BORE		ALLOWABLE DEVIATION FROM d OF SINGLE MEAN DIA., d_{mp}		ALLOWABLE DEVIATION FROM COTTER PIN DIA., d	
OVER	INCL	HIGH	LOW	HIGH	LOW
.1250	.50000	+0	-.0015	+.010	-0

ØD BASIC OUTSIDE DIA		ALLOWABLE DEVIATION FROM D OF SINGLE MEAN DIA., D_{mp}		ALLOWABLE DEVIATION FROM OUTER RING WIDTH C	
OVER	INCL	HIGH	LOW	HIGH	LOW
.4375	1.1250	+.0010	-.0005	+0	-.005

DIMENSIONS IN MILLIMETERS

Ød BASIC BORE		ALLOWABLE DEVIATION FROM d OF SINGLE MEAN DIA., d_{mp}		ALLOWABLE DEVIATION FROM COTTER PIN DIA., d	
OVER	INCL	HIGH	LOW	HIGH	LOW
3.175	12.700	+0	-.0038	+.25	-0

ØD BASIC OUTSIDE DIA		ALLOWABLE DEVIATION FROM D OF SINGLE MEAN DIA., D_{mp}		ALLOWABLE DEVIATION FROM OUTER RING WIDTH C	
OVER	INCL	HIGH	LOW	HIGH	LOW
11.112	28.575	+.025	-.013	+0	-.13

REQUIREMENTS:

MATERIAL: STEEL, MIL-S-8690, QQ-S-624, QQ-S-633, FED-STD NO. 66 STEEL NO. 50100, 51100 AND 52100. SEALS, ACETAL RESIN, NYLON, OR EQUIVALENT,. CONSTRUCTION OPTIONAL.

(B) FINISH: ROLLER OD AND SIDES CHROME PLATED IN ACCORDANCE WITH QQ-C-320, CLASS 2, EXCEPT THICKNESS TO BE .010-.018 mm (.0004-.0007 INCHES). EXPOSED SURFACES OF OTHER PARTS AS MOUNTED, ZINC-NICKEL IN ACCORDANCE WITH AMS 2417, TYPE 2, OR CADMIUM PLATED IN ACCORDANCE WITH QQ-P-416, TYPE I, CLASS 2, .0003 TO .0006 INCHES THICK.

EAXAMPLE OF PART NUMBER:

8FEE 20 A G = CAM FOLLOWER NEEDLE BEARING .500 STUD, SEALED BEARING, LUBRICATOR IN FLANGED END OF STUD, 1.250 GRIP LENGTH, 1.672 STUD LENGTH WITHOUT COTTER PIN HOLE, WITH MIL-G-81322 GREASE.

- G = MIL-G-81322 GREASE: NO CODE = MIL-G-23827 GREASE
- A = NO COTTER PIN HLOE (NOT REQUIRED FOR TYPES T OR TEE) OMIT FOR COTTER PIN HOLE
- GRIP LENGTH IN .0625 INCH INCREMENTS
- BEARING
  - F = UNSEALED WITH LUBRICATOR IN FLANGED END OF STUD
  - T = UNSEALED WITH LUBRICATOR IN THREADED END OF STUD, NO COTTER PIN HOLE
  - "-" = UNSEALED WITHOUT LUBRICATOR
  - FEE = SEALED WITH LUBRICATOR IN FLANGED END OF STUD
  - TEE = SEALED WITH LUBRICATOR IN THREADED END OF STUD, NO COTTER PIN HLOE
  - R = SEALED BEARING WITHOUT LUBRICATOR
- STUD DIAMETER IN .0625 INCH INCREMENTS

PREPARING ACTIVITY: NAVY-AS CUSTODIANS: ARMY- AT NAVY- AS AIR FORCE- 11 DLA- REVIEW: AF-84 DLA-IS USER: PROJECT NUMBER: 3110-0909	MILITARY SPECIFICATION SHEET TITLE BEARING, ROLLER, NEEDLE, TRACK ROLLER, INTERNAL STUD, TYPE VII, ANTIFRICTION, INCH	SPECIFICATION SHEET NUMBER MS21432 05 DEC 94 REV B SUPERSEDING MS2132 A 21 MARCH 1977 AMSC- N/A FSC -3110
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NOTES:

(8) 1. THE COMPLETE MS IDENTIFICATION MAY BE OBTAINED BY ADDING THE FOLLOWING SUFFIX CODE LETTERS TO THE BASIC MS NUMBER SHOWN IN THE TABULATION:

"F" DESIGNATES LUBRICATOR IN FLANGED END OF STUD.
"T" DESIGNATES LUBRICATOR IN THREADED END OF STUD AND COTTER PIN HOLE OMITTED
"A" DESIGNATES COTTER PIN HOLE OMITTED
" " DESIGNATES A SEALED BEARING.
THIRD DASH NUMBER DESIGNATES GRIP LENGTH IN 0.0625 INCH INCREMENTS (SEE EXAMPLE).

2. MS21432-3 MAY BE CODED FOR RELUBRICATION FACILITIES IN THE FLANGED END ONLY. LUBRICATION FITTING FOR THE MS21432-3 MAY BE SHIPPED SEPERATELY AND INSTALLED ON ASSEMBLY. LUBRICATORS INSTALLED IN THE THREADED END OF STUD SHALL BE FLUSH. LUBRICATORS INSTALLED IN THE FLANGED END OF STUD ON MS21432-3 AND -4 MAY EXTEND 1.3 mm (0.05 IN.) BEYOND THE FLANGED END THEREBY RECESSITATING AN INCREASE IN " " OF 1.3 mm (0.05) IN.). ON ALL OTHER SIZES THE LUBRICATOR SHALL BE FLUSH OR INDENTED.

3. THE TRACK BRINELL CAPACITY IS CRITICAL IN RESPECT TO THE ROLLER CAPACITY OF THE BREAKING. AN INCREASE IN HARDNESS OF THE TRACK WILL INCREASE THE BRINELL CAPACITY OF THE TRACK, BUT IN NO CASE SHOULD THE ROLLING CAPACITY OF THE BEARING BE EXCEEDED.

4. THE LIMIT LOAD RATING CAN BE DEFINED AS THE MAXIMUM LOAD WHICH CAN BE APPLIED TO A BEARING WITHOUT IMPAIRING THE SUBSEQUENT FUNCTIONING OF THE BEARING IN AIRFRAME APPLICATIONS.

THE ULTIMATE OR STATIC FRACTURE LOAD RATING IS NOT LESS THAN 1.5 TIMES THE LIMIT LOAD RATING.

TO FULLY REALIZE THIS RATING, COMPUTATION MUST BE PROVIDED FOR STUD DEFLECTION IN ORDER TO ASSURE FULL TRACK CONTACT UNDER LOAD.

5. FOR INSPECTION PURPOSES NOMINAL STUD LENGTH IS THE SUM OF NOMINAL GRIP AND THE THREAD LENGTH (" " DIMENSION).

6. BEARING SHALL BE PREPACKED WITH GREASE AS CODED.

7. BEARING TO BE WRAPPED IN GREASE PROOF PAPER, INDIVDUALLY PACKAGED AND MARKED WITH THE DATE OF LUBRICATION.

8. ALL DIMENSIONS TO MET AFTER PLATING.

PREPARING ACTIVITY: NAVY-AS CUSTODIANS: ARMY- AT NAVY-AS AIR FORCE- 11 DLA- REVIEW: AF-84 DLA-IS USER: PROJECT NUMBER: 3110-0909	MILITARY SPECIFICATION SHEET TITLE BEARING, ROLLER, NEEDLE, TRACK ROLLER, INTERNAL STUD, TYPE VII, ANTIFRICTION, INCH	SPECIFICATION SHEET NUMBER MS21432 05 DEC 94 REV B SUPERSEDING MS21432 A 21 MARCH 1977 AMSC- N/A FSC -3110
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